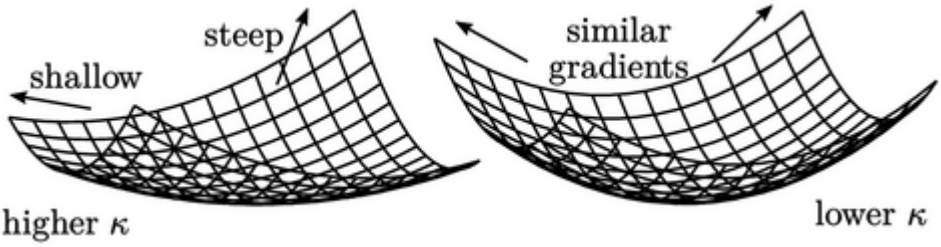
Notes on Computational Fluid Dynamics: General Principles

5.16 Preconditioning and asymmetry

The conjugate gradient (CG) method provides a good basis for a matrix solver in CFD. In theory, it provides an exact solution in N steps for a problem with N values. In practice, there is an accumulation of rounding errors from finite precision arithmetic, such that an exact solution is not attained. Problems are also too large for it to be feasible to run N steps.

CG is instead used as an iterative method, aiming to get the solution residual within an acceptable tolerance. The rate of convergence is therefore critical, which can be shown to be slower for increasing *condition number* of the matrix, $\kappa = \lambda_{\max}/\lambda_{\min}$, where $\lambda_{\max}$ and $\lambda_{\min}$ are its largest *eigenvalues*.

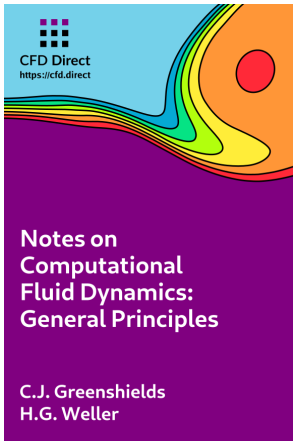


In the example with two values, the eigenvalues correspond to the gradients of the principal axes of the paraboloid described by the quadratic function. Lower κ corresponds to closer gradients, *i.e.* to a rounder "bowl" shape.

The best convergence corresponds to $\kappa = 1$ when $\mathbf{A} = \mathbf{I}$ ($\mathbf{I}$ is the identity matrix). *Preconditioning* finds a matrix $\mathbf{M}$, similar to $\mathbf{A}$, which can be inverted and multiplied to the equation as follows:

$$(\mathbf{M}^{-1} \bullet \mathbf{A}) \bullet \Psi = \mathbf{M}^{-1} \bullet \mathbf{b}. \tag{5.35}$$

The aim of preconditioning is to improve convergence by choosing a preconditioner $\mathbf{M}^{-1}$ so that $\mathbf{M}^{-1} \bullet \mathbf{A}$ is closer to $\mathbf{I}$ than $\mathbf{A}$. Effective



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methods for calculating the preconditioner $\mathbf{M}^{-1}$ are:

- diagonal-based incomplete Cholesky (DIC) for symmetric $\mathbf{A}$;
- diagonal-based incomplete LU (DILU) for an asymmetric $\mathbf{A}$.

Biconjugate gradient stabilised method

The **preconditioned CG** (PCG) method minimises the quadratic form which requires that $\mathbf{A}$ is symmetric. However, any equation containing an advection term produces an asymmetric matrix (see Sec. 5.1), when PCG is unsuitable.

The **preconditioned biconjugate gradient** (PBiCG) method is designed for asymmetric matrices. It augments the calculations of residuals and line search directions in the CG method with a second set of calculations relating to the transpose matrix $\mathbf{A}^{\mathrm{T}}$. The computational cost is approximately $2\times$ the cost of PCG.

The convergence behaviour of PBiCG is somewhat erratic, with large variations in the reduction of the residual between successive iterations. Sometimes the method breaks down and fails.

The **preconditioned biconjugate gradient stabilised** method (PBiCGStab) is a modification of PBiCG. It is exhibits much smoother convergence and robustness than PBiCG so it is far more preferable for CFD applications.

Recommended CG-based methods are summarised below.

- *symmetric*: **PCG** with **DIC** preconditioning.
- *asymmetric*: **PBiCGStab** with **DILU** preconditioning.

The implementation of preconditioning and CG-based methods can be found in: Barrett R. *et al.* *Templates for the solution of linear systems*, available at `http://www.netlib.org`.

Chris Greenshields Henry Weller 11th April 2022 Notes on CFD

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